Why ITP Longevity Drugs Work Differently in Male and Female Mice

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Abstract

Most longevity compounds tested by the NIA Interventions Testing Program work better in male mice than female mice, except rapamycin, which works better in females. We show that a mechanistic model of four coupled aging subsystems, calibrated to six male ITP results (6 free parameters, 162 frozen priors from published biology), predicts all six female outcomes from four sex-specific mechanisms in prior literature—none fitted to female data. Mean absolute error is 1.3 percentage points, a sevenfold improvement over the naive assumption that females respond like males (9.0 pp). The model separately predicts the rapamycin + acarbose combination within 0.2 pp, correctly identifying sub-additivity from shared pathway saturation. Where the model errs largest, it errs instructively: canagliflozin is underpredicted by 4 pp because the model routes the entire effect through AMPK, where females already have higher baseline activity. The canagliflozin error pattern suggests a sex-independent second pathway, consistent with FGF21/ketogenesis, and generates a falsifiable prediction: in FGF21-knockout mice, canagliflozin should lose most of its lifespan benefit.

1 Introduction

The biology of aging has attracted sustained theoretical attention for more than a century, producing a vast literature notable more for breadth than predictive power. The distinction between a theory of aging causation and a quantitative model of aging interventions is the difference between explaining *why* an organism dies and predicting *when*. Reliability theory [Gavrilov and Gavrilova, 2001] makes one piece of the problem precise: organisms are series systems that fail when any one critical subsystem reaches its threshold. The consequence is that fixing one failure mode while leaving others intact produces gains proportional to that mode's contribution and no more. The remaining modes do not vanish; they wait. This framework also predicts that drugs targeting the same subsystem will show diminishing returns in combination, while drugs targeting different subsystems should compound—a prediction that requires a multi-subsystem model to test.

The most conspicuous unexplained pattern in 20 years of ITP data is that most longevity compounds work better in males, except rapamycin, which works better in females. Jiang et al.

[Jiang et al., 2025] documented the pattern across 54 compounds and 164 trials, confirming it is systematic, not compound-specific. Garratt [Garratt, 2020] proposed qualitative mechanisms (pharmacokinetics, hormonal substrates, baseline pathway differences) but none were combined into a model that makes quantitative predictions.

Here we describe the Extensible Longevity Model (ELM): a system of ordinary differential equations that tracks four aging subsystems (mitochondrial heteroplasmy, DNA damage, epigenetic drift, and cellular senescence) coupled through NAD^+ metabolism and SASP signaling. These correspond to a subset of the Hallmarks of Aging [López-Otín et al., 2013, 2023]; the ELM attempts to put calibrated numbers on the subset that is experimentally tractable with current ITP data.

The four subsystems are not independent. They are coupled through shared molecular intermediates, most centrally NAD^+, whose availability governs DNA repair (via PARP1), mitophagy (via SIRT1/FOXO3), and epigenetic stability (via EZH2). A drug that raises NAD^+ can partially address multiple subsystems simultaneously; conversely, two drugs that both raise NAD^+ may show diminishing returns when combined.

The model has 6 free parameters (compound scale factors, calibrated to male ITP data) and 162 frozen priors imported from published biology. Four sex-specific mechanisms, each drawn from independent experimental literature, are applied to the calibrated model to generate female predictions. The female predictions use no additional fitted parameters. Their agreement, or disagreement, with observed female outcomes is the central test of whether the model has captured the right biology.

1.1 The Data Foundation

The ITP tests candidate compounds in genetically heterogeneous UM-HET3 mice at three independent sites [Miller et al., 2007, Strong et al., 2008, Harrison et al., 2009]. Six compounds show reproducible lifespan extension: rapamycin [Miller et al., 2011], acarbose [Harrison et al., 2014], 17α-estradiol [Stout et al., 2017], canagliflozin [Miller et al., 2020], aspirin [Strong et al., 2016], and glycine [Miller et al., 2019]. These span diverse mechanisms and show striking sex differences, with four of six producing male-biased or male-only effects.

The ITP subsequently tested rapamycin and acarbose in combination, reporting +34% median lifespan extension in males [Strong et al., 2022]. This combination was not used in any model fitting.

1.2 Sex Mechanisms from Prior Literature

Why do these six compounds show different sex ratios? We propose four mechanisms, each supported by prior experiments on the relevant pathway:

1. **AMPK saturation.** Females have higher baseline AMPK via estrogen-mediated LKB1 upregulation [Boudaba et al., 2018, McInnes et al., 2012], attenuating response to AMPK activators (acarbose, canagliflozin, glycine).

2. **CYP3A pharmacokinetics.** Females achieve 3.48× higher rapamycin blood levels due to reduced hepatic clearance [Lamming et al., 2014].

3. **Testosterone dependency.** 17α-estradiol requires 5α-reductase substrate [Garratt et al., 2018]; females and castrated males show zero effect.

4. **COX-2/prostacyclin interference.** Aspirin suppresses estrogen-mediated cardioprotection via COX-2$\rightarrow$PGI$_2$ [Egan et al., 2004, Grosser et al., 2006], cancelling its benefit in females.

2 Methods

2.1 Model Structure

The intuition behind the model is simple: four kinds of biological damage accumulate over the mouse's lifetime, flowing downhill into a composite biological age. The mouse dies when biological age crosses a threshold. The channels are not independent; they merge and feed into each other along the way. Heteroplasmy generates ROS, which accelerates DNA damage; DNA damage triggers senescence, which produces SASP, which drains NAD$^+$ and accelerates methylation drift. Longevity drugs work by partially blocking specific channels: rapamycin opens the autophagy drain, acarbose activates AMPK, and so on. The equations below (Equations 1–5) simply track how fast each channel fills and how much they feed into each other. The signaling intermediates (AMPK, mTORC1, SIRT1, etc.) are the valves that drugs can turn.

The ELM tracks five state variables across the mouse lifespan (Figure 1). Full parameter tables with provenance are provided in the Supplementary Material.

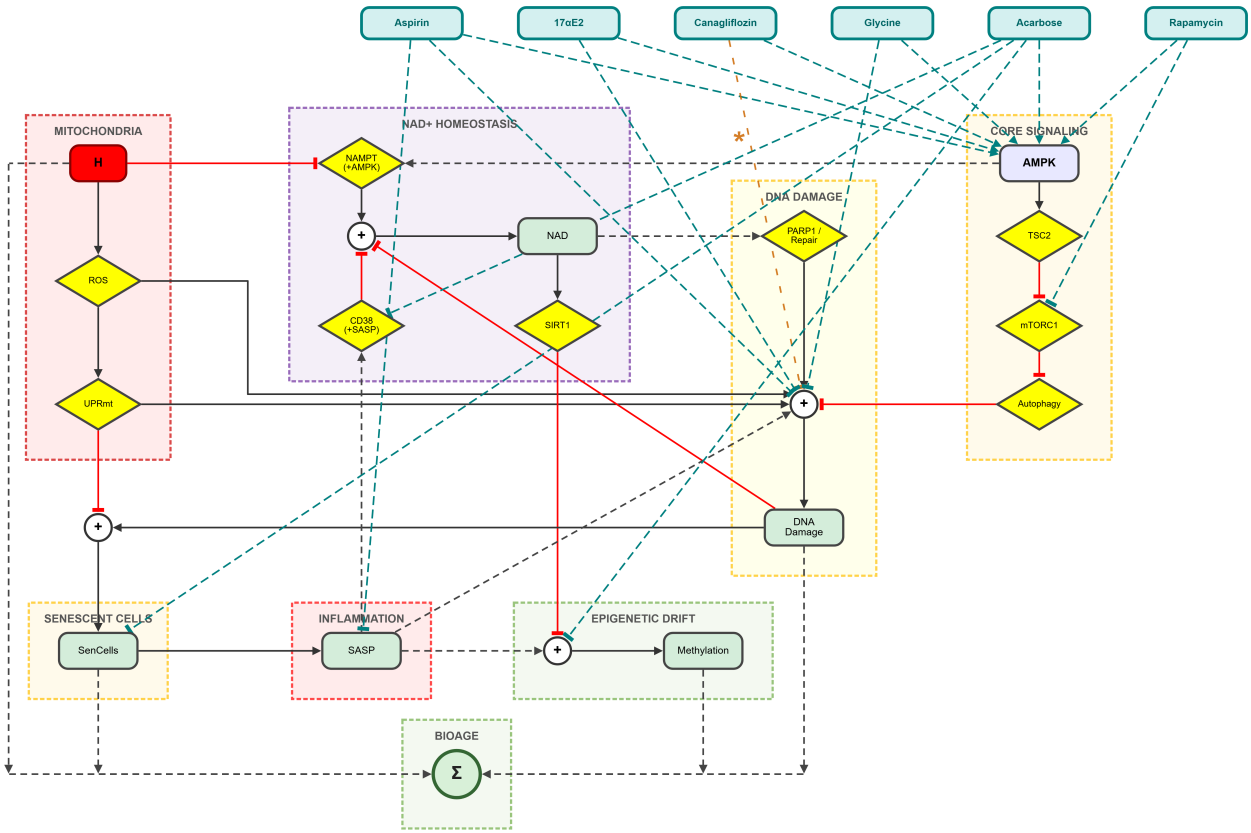

Figure 1: ELM system architecture with ITP compounds (top) and their entry points (downward dashed diagonal lines) to the aging system. Enzymes and processes (diamonds) route flows that add (black arrows) or subtract (red) from state variables (bold rectangles). Signaling (dashed arrows) affects flow rates. The orange dashed line (*) marks a hypothesized canagliflozin pathway analyzed in Section 3.1.

2.1.1 State variables

Five state variables track the aging subsystems:

- **Heteroplasmy** (H): fractional mutant mtDNA abundance. $H(0) = 0.011$, meaning 1.1% of mitochondrial DNA is mutant at birth.
- **NAD$^+$ level** (N): normalized to young-adult baseline, $N(0) = 1.0$.
- **DNA damage** (D): normalized double-strand break equivalents. $D(0) = 1.0$, meaning all damage values are measured relative to baseline at birth.
- **Senescent cell burden** (S): normalized p16$^{\text{Ink4a}}$-positive abundance. $S(0) = 1.0$; senescent cells accumulate above this baseline over the lifespan.
- **Methylation drift** (M): deviation from youthful CpG set points, $M(0) = 0$.

Ten signaling intermediates are computed from the state variables at each time step; they respond instantly, without their own differential equations (Table 1). Each is a Hill or Michaelis–Menten function of one or two inputs; for example, SIRT1 activity is $f_{\text{SIRT1}}(N) = N^{n_S}/(K_S^{n_S}+N^{n_S})$, coupling NAD^+ levels to downstream deacetylation. They serve as the entry points for compound effects and as the coupling terms between subsystems.

Table 1: Signaling intermediates: computed from state variables at each time step (no differential equations of their own).

Intermediate	Primary inputs	Function type	Key role
AMPK	energy stress, compounds	Michaelis–Menten	TSC2 activation, NAMPT boost
TSC2	AMPK	Hill ($n = 1.5$)	Rheb-GAP, mTORC1 suppression
mTORC1	TSC2, rapamycin	multiplicative	autophagy inhibition
SIRT1	NAD^+	Hill ($n = 2.0$)	FOXO3 activation, deacetylation
FOXO3	SIRT1	linear	mitophagy, stress response
Autophagy	AMPK, FOXO3	Michaelis–Menten	damage clearance
SASP	SenCells	proportional	CD38, ROS, paracrine senescence
ROS	heteroplasmy, SASP	additive	DNA damage source
EZH2	AMPK, SIRT1	inhibitory	DNMT recruitment, CpG drift
PGC-1α	AMPK, SIRT1	additive	mitochondrial biogenesis

2.1.2 Governing equations

Heteroplasmy dynamics follow the Kowald–Kirkwood replication advantage model [Kowald and Kirkwood, 1993]:

$$\frac{dH}{dt} = \underbrace{\alpha \cdot H(1-H)}_{\text{mutant replication advantage}} - \underbrace{\beta \cdot H \cdot \Phi_{\text{mito}}(t)}_{\text{mitophagy clearance}} \tag{1}$$

Mutant mtDNA outcompetes wild-type (first term) unless mitophagy removes damaged mitochondria fast enough (second term). $\Phi_{\text{mito}}(t)$ is mitophagy flux, driven by FOXO3 activity.

NAD^+ dynamics balance synthesis against three competing drains:

$$\frac{dN}{dt} = \underbrace{k_{\text{synth}} \cdot f_{\text{NAMPT}}(\text{AMPK})}_{\text{NAD}^+\text{ synthesis}} - \underbrace{k_{\text{PARP}} \cdot D \cdot N}_{\text{DNA repair consumption}} - \underbrace{k_{\text{CD38}} \cdot N}_{\text{CD38 consumption}} - \underbrace{k_{\text{SIRT}} \cdot N \cdot f_{\text{SIRT1}}(N)}_{\text{SIRT1 consumption}} \tag{2}$$

The cell makes NAD^+ (first term) but three processes consume it: repairing DNA damage, CD38 degradation, and SIRT1 deacetylation. As damage accumulates, the drains win.

DNA damage accumulates from ROS and is repaired by NAD^+-dependent pathways:

$$\frac{dD}{dt} = \underbrace{k_{\text{ROS}} \cdot (1 + \gamma_H H + \gamma_{\text{SASP}} \cdot \text{SASP})}_{\text{ROS production, amplified by heteroplasmy \& SASP}} - \underbrace{k_{\text{repair}} \cdot N \cdot D}_{\text{NAD}^+\text{-dependent repair}} \tag{3}$$

Methylation drift is accelerated by damage and SASP, restrained by AMPK-mediated suppres-

sion of EZH2, which reduces DNMT recruitment to CpG sites [Viré et al., 2006]:

$$\frac{dM}{dt} = \underbrace{k_{\text{drift}}(1 + \delta_D D + \delta_{\text{SASP}} \cdot \text{SASP})}_{\text{epigenetic drift, amplified by damage \& SASP}} - \underbrace{k_{\text{EZH2}} \cdot f_{\text{AMPK}}(\text{AMPK}) \cdot M}_{\text{AMPK-mediated maintenance}} \tag{4}$$

Senescent cell entry follows a damage-dependent Hill function plus a constant background term k_{repl} representing replicative exhaustion (Hayflick-limit arrivals per unit time); clearance is proportional to burden:

$$\frac{dS}{dt} = \underbrace{k_{\text{entry}} \cdot \frac{D^n}{K_D^n + D^n}}_{\text{damage-driven entry}} + \underbrace{k_{\text{repl}}}_{\text{replicative exhaustion}} - \underbrace{k_{\text{clear}} \cdot S}_{\text{immune clearance}} \tag{5}$$

2.1.3 Biological age and survival

The four damage channels feed into a single composite biological age $B(t)$. NAD^+ is excluded because it acts as a coupling intermediate rather than an independent aging endpoint; its effects are captured through its influence on DNA repair, methylation, and mitophagy:

$$B(t) = \left(\frac{w_H}{\hat{H}_0}\right)H \; + \; \left(\frac{w_D}{\hat{D}_0}\right)D \; + \; \left(\frac{w_M}{\hat{M}_0}\right)M \; + \; \left(\frac{w_S}{\hat{S}_0}\right)S \tag{6}$$

with equal weights $w_H = w_D = w_M = w_S = 0.25$, adopted as an agnostic default rather than inferred from data, where $\hat{H}_0, \hat{D}_0, \hat{M}_0, \hat{S}_0$ are normalization constants derived from control simulation values at $t = 1.0$ (median lifespan), ensuring $B(1.0) = 1.0$ by construction. Biological age determines the death rate through a Gompertz function: $h(t) = h_0 \exp(\lambda \cdot B(t))$, calibrated to the UM-HET3 control survival curve. We define lifespan extension as the percent change in median survival between treated and control simulations.

The equal-prior weight assignment reflects agnosticism about which subsystem dominates mouse mortality and avoids consuming degrees of freedom on weight tuning. We checked this assumption against three published biological age clocks (GrimAge [Lu et al., 2019], PhenoAge [Levine et al., 2018], DunedinPACE [Belsky et al., 2022]) by classifying each clock's biomarkers into ELM subsystems and summing weights per category (Supplementary Table S13). The clock decomposition does not identify a unique weight vector, but it disfavors extreme ones: setting $w_M = 0.50$, for example, is outside the range implied by both GrimAge (20–30%) and PhenoAge (5–15%). All four equal weights (0.25) fall within every clock's implied range. Sensitivity analysis (Section 3.3) confirms that predictions are stable across the full plausible range.

2.1.4 Compound implementation

Each compound perturbs one or more pathway rate constants (Table 2). An injection value of 0.20 on a pathway means the compound shifts that pathway's activity by 20% of its dynamic range, from its untreated baseline toward its theoretical maximum effect. A value of 0.42 for rapamycin's

mTORC1 inhibition, for example, means rapamycin at the ITP dose pushes mTORC1 42% of the way from its baseline toward full inhibition.

These values are not set by hand. For each compound, the pharmacological literature establishes which pathways the drug engages. The pathway ratios, how the drug's total effect is divided among its pathways, are estimated from published mechanism-of-action studies: the dominant target receives the largest share. Rapamycin, for example, is assigned 60% to direct mTORC1 inhibition (its primary target) and 40% to AMPK activation (a secondary feedback effect). These ratios reflect mechanism hierarchy rather than direct quantitative decomposition. They are constrained assumptions, not fitted parameters and not direct measurements. Exact values are documented in Supplementary Table S2.

A single scale factor α_c then sets the overall magnitude: one free parameter per compound, adjusted until the model matches the observed male ITP lifespan extension. The injection value for each pathway equals its share of the split times α_c. The resulting values range from 0.03 to 0.42; if the model required most values near 0 or near 1, the pathway assignments would be suspect. The one-at-a-time sensitivity analysis (Supplementary Section S2.2) confirms that individual ratio perturbations of ±20% shift predicted lifespan by less than 0.1 pp, so the exact splits do not drive the results.

Table 2: Compound pathway assignments and calibrated injection values. For each compound, the pathway *ratios* (column 4) are estimated from mechanism-of-action studies; they encode which pathways the drug engages and in what approximate proportion (Supplementary Table S2). The *scale factor* α_c (column 5) is the single free parameter per compound, found by root-finding to the male ITP target. The *injection value* (column 6) equals ratio × α_c. Six compounds, six scale factors, six data points.

Compound	Mechanism	Pathway	Ratio (lit.)	α_c (fitted)	Injection
Rapamycin	mTORC1 inhib.	mTORC1	0.60	0.70	0.42
		AMPK	0.40		0.28
Acarbose	α-glucosidase	AMPK	0.50	0.34	0.17
		gut microbiome	0.50		0.17
17α-Estradiol	5α-reductase	AMPK	0.50	0.35	0.18
		antioxidant	0.50		0.18
Canagliflozin†	SGLT2 inhib.	AMPK	1.00	0.16	0.16
Aspirin	COX inhib.	antioxidant	0.50	0.20	0.10
		AMPK	0.30		0.06
		anti-inflam.	0.20		0.04
Glycine	antioxidant	antioxidant	0.90	0.32	0.28
		AMPK	0.10		0.03

†Dual-action decomposition (antioxidant 0.88 / AMPK 0.12, $\alpha_c = 0.60$) explored in Section 3.1.

Three pathways in Table 2 act by modifying rate constants within the existing equations rather than adding new ones. **Antioxidant** reduces the ROS level in Equation 3: k_{ROS} is multiplied by a protection factor $(1 - k_{\mathrm{aox}} \cdot \mathrm{antioxidant})$, so glycine's large antioxidant injection (0.28) directly sup-

presses oxidative DNA damage. **Gut microbiome** reduces CD38 activity in Equation 2: acarbose's gut fermentation products lower CD38-mediated NAD^+ consumption, slowing the $k_{\mathrm{CD38}} \cdot N$ drain term. **Anti-inflammatory** accelerates SASP clearance, reducing the SASP-driven amplification terms that appear in Equations 2, 3, and 4. All three pathways modify rate constants within the existing ODE structure rather than adding new equations.

2.2 Calibration

We calibrate the model with one free parameter per compound: a scale factor α_c that we adjust (via root-finding) until the predicted male lifespan extension exactly matches the ITP observation:

$$\Delta L_{\mathrm{pred}}^{(c,\mathrm{male})} = \Delta L_{\mathrm{ITP}}^{(c,\mathrm{male})} \tag{7}$$

We calibrate to male targets because all six compounds produce non-zero male effects, providing six independent constraints. Only four compounds produce non-zero female effects (17α-estradiol and aspirin show zero female benefit), so female calibration would leave two scale factors undetermined. The calibration direction reflects data availability, not biological primacy.

Male predictions are exact by construction (error < 0.01 pp). An additional 162 quantities are fixed before calibration: 115 taken from published measurements, 24 set within biologically plausible ranges (including 6 adjusted to produce realistic untreated-mouse biology), and 23 structural constants (normalization, coupling topology). None were adjusted to match ITP intervention outcomes. Full provenance is documented in the Supplementary Material.

We then compute female predictions using the same scale factors plus the four sex-specific mechanisms. No parameters are adjusted to match female data. These are genuine out-of-sample predictions.

2.3 Sex-Specific Implementation

AMPK saturation. Female AMPK baseline is $b = 1.3\times$ male [Boudaba et al., 2018, McInnes et al., 2012]. We represent this attenuation with the saturating form $f_{\mathrm{atten}} = 1/(1 + k(b - 1))$; with $b = 1.3$ and $k = 8$, this gives 0.29, meaning females get roughly one-third of the AMPK-mediated benefit that males get. This single functional form governs female response for all three AMPK-activating compounds (acarbose, canagliflozin, glycine). Non-AMPK pathway injections (antioxidant, gut microbiome) pass through the sex modifier at full strength.

CYP3A pharmacokinetics. Female effective rapamycin exposure is $3.48\times$ male [Lamming et al., 2014], applied via Hill-function mTORC1 inhibition.

Testosterone dependency. 17α-estradiol perturbation is set to zero in female simulations [Garratt et al., 2018].

COX-2 interference. A PGI_2 offset cancels aspirin's antioxidant benefit in females [Egan et al., 2004, Grosser et al., 2006].

2.4 Combination Simulation

In combination simulations, we activate both compound perturbations at once, each applied to its own rate constants. Cross-compound interactions arise only through shared state variables. The rapamycin + acarbose combination was simulated from a 9-month start time matching the ITP protocol [Strong et al., 2022].

2.5 Sensitivity Analysis

We varied each biological age weight across $[0.10, 0.50]$, redistributing the remainder equally, and recalibrated the model at each value. We independently perturbed each compound scale factor $\pm 20\%$.[1]

3 Results

3.1 Sex Differences: Four Mechanisms, Zero Sex-Specific Fitting

Using four sex-specific mechanisms drawn from prior literature and fixed before comparison to female ITP outcomes, the model predicts the six female outcomes, with 1.3 pp mean absolute error (four of six within 1 pp; Table 3). This is the primary result.

Table 3: ITP observed vs. ELM predicted lifespan extension. Male predictions are calibrated (6 free parameters); female predictions in the main model are out-of-sample tests. †The daggered canagliflozin entry shows the post hoc dual-action variant ($f = 0.88$; Section 3.1, Figure 3) and is not out-of-sample.

	Male (calibrated)		**Female (predicted)**		
Compound	ITP	ELM	ITP	ELM	Sex mechanism
Rapamycin	+23.0%	+23.0%	+26.0%	+26.7%	CYP3A pharmacokinetics
Acarbose	+22.0%	+22.0%	+5.0%	+7.6%	AMPK saturation
17α-Estradiol	+19.0%	+19.0%	0%	0%	testosterone dependency
Canagliflozin	+14.0%	+14.0%	+9.0%	+5.0%†	AMPK saturation
Aspirin	+8.0%	+8.0%	0%	0%	estrogen interference
Glycine	+6.0%	+6.0%	+4.0%	+3.8%	AMPK saturation

Uncertainty estimates. ITP studies do not report confidence intervals on median extension. We estimate ± 2.5 pp (1 SE) per compound per sex, based on: ITP sample sizes of $\sim$130–150 treated and $\sim$270 control mice per sex [Miller et al., 2007], bootstrap SE of the median ($1.253 \times \sigma/\sqrt{n}$, with $\sigma \approx 170$ days for UM-HET3 lifespans), and propagation through the treated-minus-control ratio. This is consistent with the ITP's stated power to detect a 10% change in mean lifespan at 80% power. Model prediction uncertainty is computed by one-at-a-time perturbation of all 141 frozen parameters

[1] ODEs were integrated with SciPy `solve_ivp` (RK45). All code is available at `https://github.com/silmonbiggs/extensible-longevity-model` (MIT license).

(±10%), with male scale factors re-calibrated at each perturbation (Supplementary §S2.2); per-compound female sensitivities are then scaled by each parameter's plausible range and summed in quadrature (RSS). The resulting model SE ranges from 0 pp (aspirin, 17α-estradiol: categorical mechanisms) to 2.3 pp (rapamycin: PK-mediated sensitivity to heteroplasmy parameters).

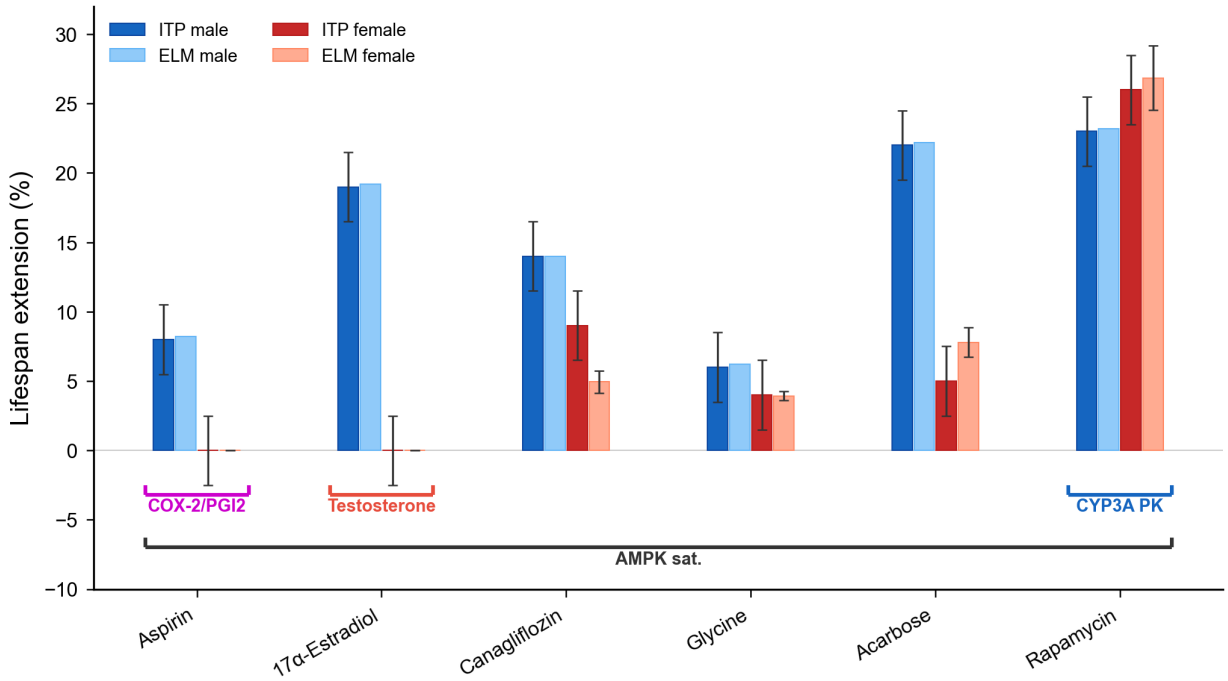

Figure 2: ELM predictions vs. ITP observations for all six compounds, both sexes (pure-AMPK canagliflozin model). For each compound, paired bars show ITP observed (dark) and ELM predicted (light), with blue for males and red for females. Error bars on ITP bars: estimated SE of the median extension (±2.5 pp; see text). Error bars on ELM female bars: model parameter uncertainty (OAT-derived RSS; Supplementary §S2.2). ELM male bars carry no error bars because each male prediction is solved exactly to its ITP target (one free parameter per compound). Four of six female predictions fall within combined 1σ uncertainty; canagliflozin and acarbose exceed it.

AMPK saturation (acarbose, canagliflozin, glycine). A 30% higher AMPK baseline, on a steeply saturating curve, leaves only 29% of the male drug response available (attenuation factor 0.29). The per-compound predictions: glycine is predicted well (+3.8% vs. +4.0%) because 90% of its injection is antioxidant, which passes through the sex modifier unattenuated. Acarbose is overpredicted (+7.6% vs. +5.0%, 2.6 pp). Canagliflozin is underpredicted (+5.0% vs. +9.0%, 4.0 pp).

The opposite-sign errors for acarbose and canagliflozin are informative. If both compounds acted purely through AMPK, their female attenuation should be similar. Instead they err in opposite directions: acarbose is overpredicted, canagliflozin is underpredicted. The model suggests a natural explanation: the AMPK saturation mechanism diminishes only the `ampk` and `gut_microbiome` pathway injections; an antioxidant fraction passes through at full strength for females. If part

of canagliflozin's effect operates through a sex-independent pathway, females would retain that component unattenuated, raising the predicted female benefit toward the observed value.

Three independent lines of evidence support a sex-independent second pathway: SGLT2 inhibition shifts substrate utilization toward fatty acid oxidation and ketogenesis [Ferrannini et al., 2016]; canagliflozin reprograms systemic metabolism via FGF21-dependent pathways in mice [Osataphan et al., 2019]; and SGLT2 inhibitors activate ketogenic starvation sensors independently of glucose lowering [Packer, 2020]. In the model, this maps to the antioxidant node, which reduces ROS-driven DNA damage—precisely the downstream endpoint of Nrf2 activation.

To quantify this hypothesis, we swept the antioxidant fraction f from 0 to 1, recalibrating the male scale factor at each point via root-finding so that the male prediction remains exactly +14% regardless of the split (Figure 3). This introduces one additional parameter, the pathway split f, constrained by one datum: female canagliflozin = +9%. The revised fit is therefore not out-of-sample, but it yields a new falsifiable prediction.

The sweep constrains f but cannot pin it down precisely. The mean female error (Figure 3B, black) stays below the expected MAE of a perfect model measured against ITP-precision data ($\sigma\sqrt{2/\pi} \approx 2.0$ pp) for the *entire* sweep, so the mean-error minimum at $f^* = 0.88$ is a point estimate within the noise, not a statistically resolved optimum. The maximum single-compound error (red) provides the binding constraint: it drops below the ITP measurement SE (2.5 pp) for $f \in [0.70, 0.96]$ and flatlines at 2.8 pp across that range. In this band, canagliflozin is no longer the worst-predicted compound—acarbose is (+7.8% predicted vs. +5.0% observed), and no further improvement to the maximum error is possible without addressing acarbose independently.

What the data support is therefore: a dominant sex-independent pathway (f in the range 0.70–0.96), with the AMPK-mediated fraction as a minor contributor. At the point estimate $f^* = 0.88$, the canagliflozin prediction improves from +5.0% to +8.8% (0.2 pp error) and mean female MAE drops from 1.3 to 0.65 pp.

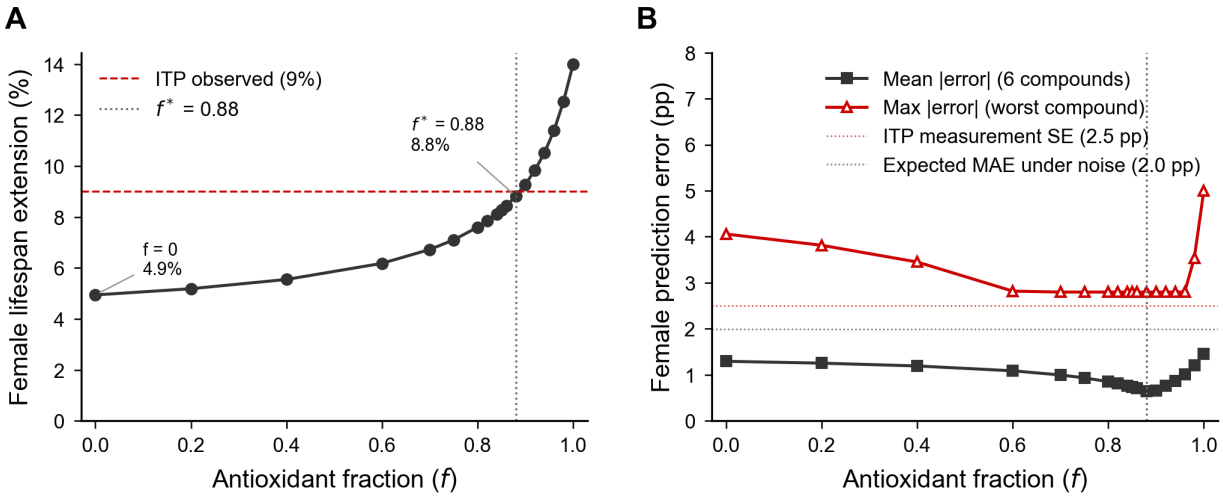

Figure 3: Canagliflozin dual-action sweep. **(A)** Female lifespan extension as a function of the antioxidant fraction f, with male extension recalibrated to +14% at each point. The dashed line marks the ITP observed value (+9%). **(B)** Mean and maximum female prediction error across all six ITP compounds. Dotted reference lines show the ITP measurement SE (2.5 pp, red) and the expected MAE of a perfect model measured against data of this precision ($\sigma\sqrt{2/\pi} \approx 2.0$ pp, grey). The mean error is below its noise floor for all f; the maximum error constrains f to $[0.70, 0.96]$.

The falsifiable prediction: in FGF21-knockout mice, canagliflozin should retain at most the AMPK-mediated fraction of its lifespan benefit—a residual male extension of 1–4 pp (instead of +14%), and near-zero female extension (the small AMPK residual is further diminished by AMPK saturation). Plasma FGF21 and β-hydroxybutyrate in canagliflozin-treated wild-type mice should be elevated to a similar degree in both sexes, confirming the sex-independent nature of the dominant pathway.

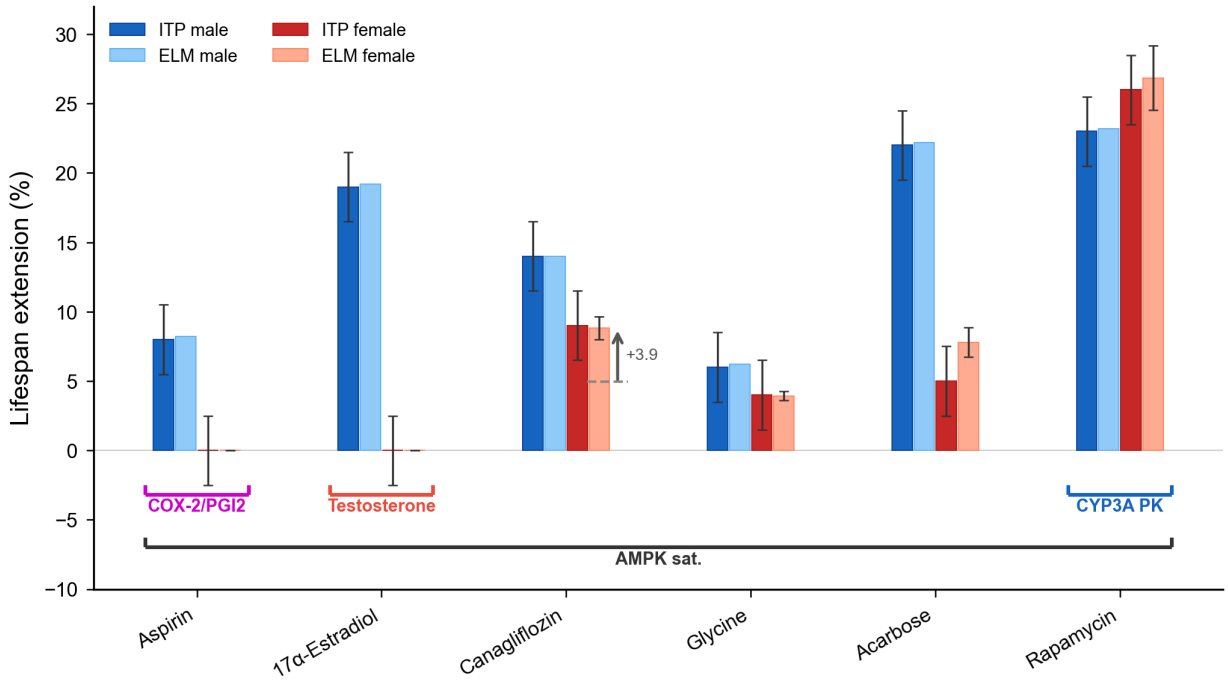

Figure 4: Same layout as Figure 2, with canagliflozin updated to the dual-action model ($f^* = 0.88$). The canagliflozin female gap closes from 4.1 pp to 0.2 pp. Five of six compounds now fall within combined 1σ uncertainty; acarbose remains the sole outlier.

Pharmacokinetics (rapamycin). Rapamycin is the one compound showing greater female efficacy (+26% F vs. +23% M). The AMPK saturation mechanism predicts the opposite. The reversal is pharmacokinetic: female mice achieve 3.48-fold higher blood rapamycin levels at equivalent dietary doses due to reduced CYP3A hepatic clearance [Lamming et al., 2014], more than offsetting the AMPK disadvantage. The model predicts +26.7% female, an error of 0.7 pp.

Testosterone dependency (17α-estradiol). Without testosterone substrate, the mechanism is inoperative in females, confirmed experimentally in castrated males, which lose efficacy entirely [Garratt et al., 2018]. Predicted: 0%. Observed: 0%.

Estrogen interference (aspirin). COX inhibition suppresses the COX-2→PGI_2 pathway through which estrogen mediates cardioprotection [Egan et al., 2004, Grosser et al., 2006]. The net effect in females neutralizes a protective pathway, offsetting aspirin's antioxidant and anti-inflammatory benefits. Predicted: 0%. Observed: 0%.

No sex-specific parameters were adjusted to ITP lifespan data. The AMPK baseline ratio (1.3×) and rapamycin PK ratio (3.48×) come from published measurements; the saturation curvature ($k = 8$) is a round number set near the midpoint of the plausible range [3–20] (Section 4.2). All three were fixed before examining female ITP outcomes. For comparison, a naive null model that assumes female extension equals male extension produces a mean absolute error of 9.0 pp—the four sex mechanisms reduce this sevenfold, to 1.3 pp.

Figure 5 shows predicted survival curves for rapamycin-treated and control mice in both sexes.

In males, rapamycin shifts the survival curve rightward by a modest margin; the two curves separate gradually. In females, the separation is visibly larger: higher effective drug exposure (3.48× blood levels) pushes the treated curve further right, producing a larger gap at every survival quantile. The model captures this asymmetry from the CYP3A pharmacokinetic difference alone, without any female-specific fitting.

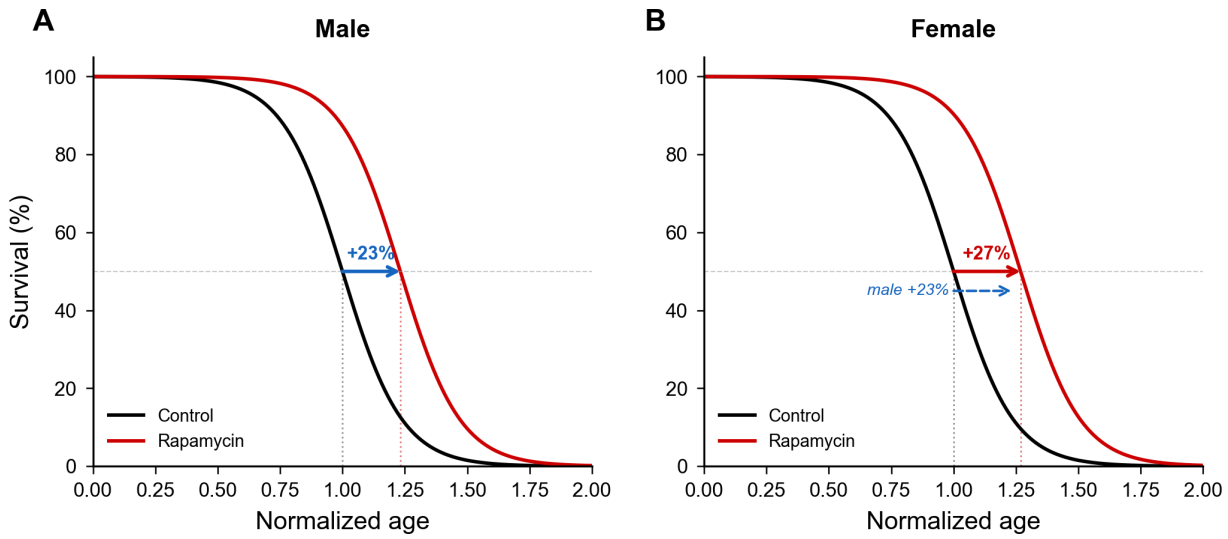

Figure 5: Predicted survival curves for rapamycin-treated (red) vs. control (black) mice. Left: males (+23%). Right: females (+26.7%). The larger female separation reflects 3.48× higher effective drug exposure from reduced CYP3A hepatic clearance.

3.2 Out-of-Sample Validation: Rapamycin + Acarbose

The ELM predicts +34% for the rapamycin + acarbose combination (male, 9-month start); the ITP observed +34% [Strong et al., 2022]. This combination was not used in any parameter fitting: neither the compound scale factors (calibrated to individual male targets) nor the biological age weights (equal priors, fixed before calibration) were adjusted to match this result.

The prediction is not trivially correct. Both compounds activate AMPK, but through different routes: rapamycin indirectly (mTORC1 inhibition relieves negative feedback on AMPK) and acarbose directly (energy stress from reduced carbohydrate absorption). Both converge on the same downstream autophagy pathway, which follows Michaelis–Menten kinetics with $K_a = 0.35$.

At ITP doses, rapamycin alone drives autophagy to ∼78% of maximum; acarbose drives it to ∼65%. Combined, the pathway encounters diminishing returns near saturation. A naive additive model predicts +45%. The ELM predicts +34%. The ITP observes +34% (Figure 6). The naive model is not close. The ELM identifies the direction of the interaction (sub-additive), its approximate magnitude, and the correct mechanistic reason: pathway saturation at a shared node.

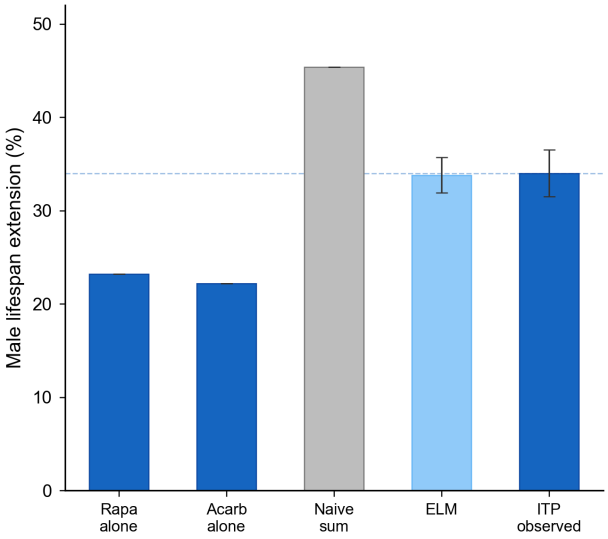

Figure 6: Out-of-sample combination prediction: rapamycin + acarbose (male, 9-month start). Individual compound effects and ITP observed in dark blue (data); naive additive sum in gray; ELM prediction in light blue (model). Error bars: ±1.9 pp model SE (ELM), ±2.5 pp estimated SE (ITP). The naive sum (+45%) overpredicts by 11 pp; the ELM (+34%) matches the ITP (+34%) within combined uncertainty.

3.3 Sensitivity and Robustness

Weight sensitivity. We swept the methylation weight across $[0.05, 0.50]$, well beyond the biologically plausible clock-decomposition range of $[0.05, 0.30]$ (Supplementary Section S3). The data cannot strongly constrain this weight: the mean calibration error stays below the ITP measurement noise floor (∼2.5 pp SE) everywhere, and the maximum single-compound error barely exceeds it (Figure 7A). Equal 25% weights remain compatible with the data on hand. The out-of-sample combination prediction provides mild additional constraint: the predicted rapa+acarbose extension crosses the observed +34% near $w_M = 0.25$ and remains within ±2.5 pp (one ITP SE) for $w_M \in [0.10, 0.40]$ (Figure 7B).

Parameter perturbation. Perturbing each compound scale factor ±20% shifts individual predictions by at most 2 pp. No perturbation reverses any sex-difference direction or changes the qualitative combination prediction.

Background aging rate test. If the model were just rescaling a single aging rate, then changing any one parameter would shift all predictions by the same fraction. We tested this by increasing the heteroplasmy replication advantage by 10%: all 12 predictions shift, but by different amounts (ratios ranging from 0.151 to 0.207, a 37% spread). The model is not a disguised one-parameter fit.

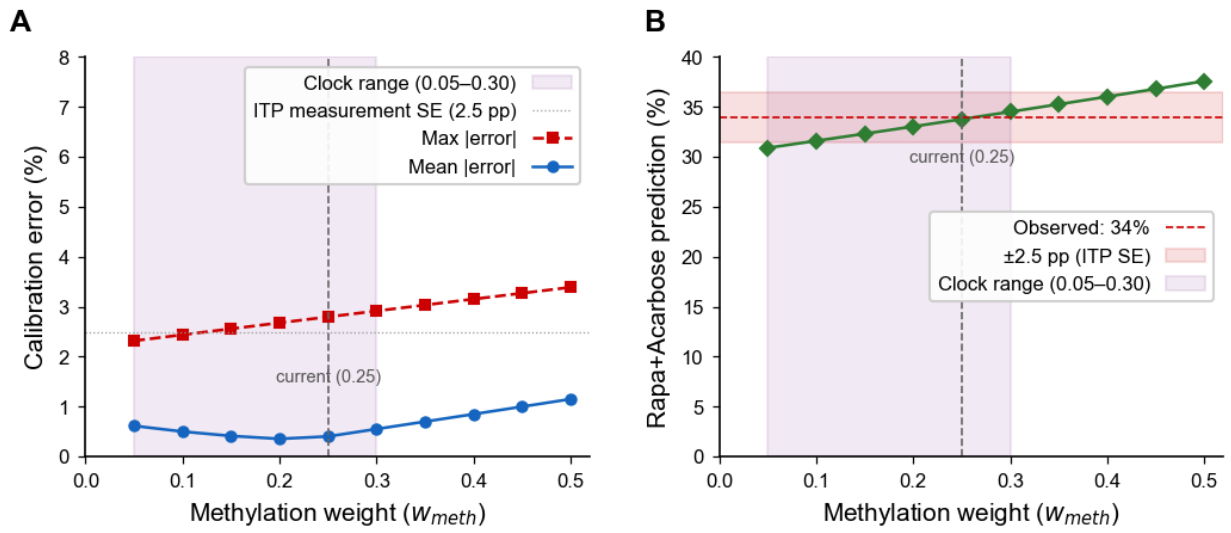

Figure 7: Biological age weight sensitivity. **(A)** Mean and maximum calibration error across all 12 compound×sex predictions as the methylation weight varies from 0.05 to 0.50. The dashed line marks the ITP measurement SE (∼2.5 pp); mean error stays well below this noise floor across the entire range. **(B)** Out-of-sample rapa+acarbose combination prediction. The salmon band shows ±2.5 pp (one ITP SE) around the observed +34%. The prediction falls within this band for $w_M \in [0.10, 0.40]$. Purple shading marks the biologically plausible clock-decomposition range.

4 Discussion

4.1 The Sex Difference Explanation

No single mechanism here is new; each has been proposed before. What is new is that four independent mechanisms, drawn from four different literatures, all produce correct predictions when combined in the same quantitative model. That all four produce the correct directional predictions, with four of six female outcomes within 1 pp, supports the claim that the model captures biologically relevant structure. If the canagliflozin pathway split is accepted, five of six fall within 1 pp, with acarbose (+7.6% vs. +5.0%, 2.6 pp) the sole remaining outlier.

4.2 The Combination Validation

The rapamycin + acarbose prediction (+34% vs. +34%) provides supporting evidence beyond the sex differences. Its value lies in what the model had to get right: predicting sub-additivity rather than the naive +45%, for the correct mechanistic reason. But one combination prediction does not validate the full model; it validates the specific pathway saturation mechanism that both compounds share. The harder test, cross-pathway interactions involving mechanistically independent axes, requires multi-compound experimental designs that have not yet been conducted, because single-compound testing, however rigorously conducted, cannot see what is behind a wall it has not yet removed.

4.3 What the Data Constrains

The model has 6 calibrated parameters and 162 frozen priors: 115 from published measurements, 24 set within biologically plausible ranges, and 23 structural constants. It is a 6-parameter model that has imported a large body of experimental biology as fixed assumptions. The model's predictive success depends on a large imported prior structure, not on six calibrated parameters alone. Those priors, whether measured directly, constrained by published biology, or set at plausible midpoints, are the best information presently available. But the priors are not all equally right. The first failures are likely to arise where the imported biological assumptions are weakest or incomplete. One-at-a-time sensitivity analysis of all 141 perturbable frozen priors is provided in the Supplementary Material (§S2.2).

4.4 Downstream Predictions

Beyond the sex differences and combination validation reported here, the model generates predictions about multi-compound interventions. The most striking: when the four best ITP compounds are combined, the model predicts that mitochondrial heteroplasmy becomes the dominant remaining cause of aging-related death—rising from 25% to 40% of biological age at death. The reason is simple: no ITP compound directly targets mitochondrial DNA quality control. The remaining modes do not vanish; they wait. This implies that the next large gain requires addressing a different aging axis entirely. These predictions, with novel target rankings and specified falsification conditions, are reported separately (Biggs 2025, preprint).

4.5 Limitations

Cross-pathway independence. The central unvalidated assumption is that the four aging axes compound approximately independently in living animals. The ODE coupling structure encodes known interactions (SASP→NAD^+ drain, ROS→DNA damage, AMPK→EZH2 methylation), but the model assumes these are the dominant cross-pathway interactions. Higher-order feedback not captured in the system could invalidate combination predictions.

AMPK saturation parameter. The diminishing-returns curvature $k = 8$ is a modeling choice, not a directly measured quantity. It is set as the midpoint of the plausible range for hyperbolic saturation of the LKB1→AMPK→TSC2 cascade. Different values of k shift the AMPK-compound female predictions in concert. The canagliflozin dual-action split ($f^* = 0.88$) was determined at $k = 8$; a different k would shift the optimal f^* but preserve the qualitative conclusion that the dominant canagliflozin pathway is sex-independent.

Scope. The model does not include Wnt/β-catenin, Notch, telomere dynamics, immune senescence, AGE crosslink accumulation, or organ-specific fibrosis. It makes no quantitative claims about human outcomes. These are targets for future extension of the model. The ODE architecture is modular: each subsystem couples to the others only through named signaling intermediates, so new subsystems can be added without refitting the existing structure, as long as the new module says

how it connects to the rest.

4.6 Relationship to Existing Models

No individual equation in ELM is new; each draws on reliability theory [Gavrilov and Gavrilova, 2001], mechanistic aging models [Kowald and Kirkwood, 1993, Kogan et al., 2015, Podolskiy et al., 2015], and the hallmarks framework [López-Otín et al., 2013, 2023]. What is new is the combination: calibration to ITP lifespan data, prospective out-of-sample validation against ITP sex data, and specified falsification conditions. To our knowledge, prior models in this space have not been calibrated to ITP data or validated prospectively on unseen outcomes, which raises a question about what, exactly, they are predicting. A model that cannot be wrong is not a model; it is a metaphor.

4.7 Conclusions

A 6-parameter model, calibrated only to male ITP data, predicts all six female outcomes with 1.3 pp mean absolute error (four of six within 1 pp) using four independently motivated sex mechanisms—and predicts the rapamycin + acarbose combination within 0.2 pp, for the correct mechanistic reason. Where it errs, the pattern of error suggests a second, sex-independent pathway, which the model quantifies ($f^* = 0.88$, consistent with FGF21/ketogenesis) and makes falsifiable (FGF21-knockout males: 1–4 pp residual extension vs. +14% in wild type).

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